

GUILIN, China

WMO: 579570

Lat: 25.3226N

Lon: 110.3046E

Elev: 545

StdP: 14.41

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>33.9 | (c)<br>36.1 | (d)<br>17.9                   | (e)<br>13.9 | (f)<br>48.6 | (g)<br>22.8 | (h)<br>17.5 | (i)<br>47.9 | (j)<br>17.4           | (k)<br>46.6 | (l)<br>15.8 | (m)<br>45.3 | (n)<br>5.2            | (o)<br>20 | (p)<br>0.346 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>14.0            | (c)<br>95.4     | (d)<br>78.1 | (e)<br>93.6 | (f)<br>77.9 | (g)<br>91.8 | (h)<br>77.7 | (i)<br>81.0         | (j)<br>89.0 | (k)<br>80.2 | (l)<br>88.0 | (m)<br>79.6 | (n)<br>87.0 | (o)<br>6.2           | (p)<br>180 |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 79.1                            | 154.3 | 84.0 | 78.6 | 151.7 | 83.7 | 77.5 | 145.7 | 82.7 | 45.5          | 88.7 | 44.6 | 88.9 | 43.8 | 87.3 | 85.6              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|-------|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |       |
| Min               |      | Max  |                            | Min  | Max  | Min                | Max                                                | Min       | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 15.9              | 13.8 | 12.0 | DB                         | 30.5 | 99.2 | 2.0                | 2.0                                                | 29.0      | 100.7 | 27.8       | 101.9 | 26.7       | 103.0 | 25.2       | 104.5 |
|                   |      |      | WB                         | 28.8 | 83.0 | 1.9                | 1.4                                                | 27.4      | 84.0  | 26.3       | 84.8  | 25.2       | 85.6  | 23.8       | 86.6  |

Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 67.6   | 47.6  | 51.4  | 57.8  | 67.8  | 75.1  | 80.3  | 83.4  | 83.5  | 79.2  | 71.3  | 61.4  | 51.5  | (6)  |
|                                                                        | DBStd       | 13.95  | 6.92  | 8.49  | 7.48  | 6.90  | 4.84  | 3.78  | 3.13  | 3.40  | 5.03  | 5.58  | 6.63  | 5.81  | (7)  |
|                                                                        | HDD50       | 278    | 130   | 76    | 17    | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 4     | 50    | (8)  |
|                                                                        | HDD65       | 1804   | 540   | 388   | 247   | 48    | 1     | 0     | 0     | 0     | 1     | 14    | 145   | 420   | (9)  |
|                                                                        | CDD50       | 6698   | 56    | 116   | 258   | 534   | 778   | 908   | 1035  | 1037  | 875   | 660   | 345   | 96    | (10) |
|                                                                        | CDD65       | 2747   | 0     | 8     | 22    | 132   | 314   | 458   | 570   | 572   | 425   | 209   | 36    | 1     | (11) |
|                                                                        | CDH74       | 26073  | 2     | 42    | 105   | 751   | 2232  | 4284  | 6474  | 6482  | 4085  | 1391  | 219   | 6     | (12) |
|                                                                        | CDH80       | 10199  | 0     | 6     | 13    | 164   | 690   | 1502  | 2832  | 2896  | 1662  | 404   | 30    | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 4.8    | 5.2   | 5.4   | 4.6   | 4.5   | 4.3   | 4.0   | 4.2   | 4.4   | 5.0   | 5.3   | 4.9   | 5.2   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 72.0   | 2.2   | 3.3   | 5.2   | 10.0  | 13.3  | 11.9  | 8.1   | 6.3   | 2.8   | 3.7   | 3.3   | 1.8   | (15) |
|                                                                        | PrecMax     | 105.5  | 5.4   | 8.1   | 12.8  | 19.8  | 31.4  | 28.5  | 23.2  | 14.5  | 10.1  | 11.3  | 9.7   | 4.6   | (16) |
|                                                                        | PrecMin     | 50.7   | 0.0   | 0.5   | 1.3   | 3.0   | 3.0   | 3.5   | 1.6   | 0.6   | 0.0   | 0.1   | 0.0   | 0.0   | (17) |
|                                                                        | PrecStd     | 12.4   | 1.5   | 1.8   | 2.4   | 4.2   | 5.5   | 5.9   | 5.3   | 3.4   | 2.0   | 3.0   | 2.7   | 1.3   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 71.2  | 79.1  | 82.1  | 88.0  | 91.8  | 94.2  | 97.9  | 98.5  | 95.8  | 91.4  | 83.9  | 73.4  | (19) |
|                                                                        |             | MCWB   | 60.9  | 66.1  | 70.5  | 74.0  | 76.2  | 78.3  | 78.7  | 78.7  | 76.7  | 72.2  | 68.0  | 58.9  | (20) |
|                                                                        | 2%          | DB     | 66.1  | 73.5  | 76.9  | 84.4  | 89.4  | 91.7  | 95.2  | 95.8  | 93.4  | 87.5  | 79.7  | 68.3  | (21) |
|                                                                        |             | MCWB   | 55.6  | 63.5  | 66.5  | 71.9  | 74.9  | 77.7  | 78.6  | 78.3  | 76.4  | 71.1  | 64.3  | 54.6  | (22) |
|                                                                        | 5%          | DB     | 61.9  | 69.3  | 73.2  | 81.2  | 86.7  | 89.8  | 93.3  | 93.6  | 91.3  | 84.4  | 76.3  | 64.7  | (23) |
|                                                                        |             | MCWB   | 53.9  | 62.0  | 64.3  | 71.3  | 74.5  | 77.7  | 78.7  | 78.1  | 75.9  | 68.9  | 63.3  | 53.4  | (24) |
|                                                                        | 10%         | DB     | 58.1  | 63.9  | 69.0  | 78.6  | 84.1  | 87.7  | 91.3  | 91.6  | 88.9  | 81.7  | 72.6  | 61.2  | (25) |
|                                                                        |             | MCWB   | 51.5  | 58.1  | 62.2  | 70.2  | 73.8  | 77.5  | 78.7  | 78.1  | 75.2  | 67.8  | 62.2  | 52.3  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 63.4  | 68.6  | 72.2  | 76.6  | 80.0  | 81.6  | 82.4  | 82.1  | 80.5  | 76.9  | 72.4  | 64.9  | (27) |
|                                                                        |             | MCDB   | 68.0  | 76.5  | 79.1  | 83.5  | 86.9  | 87.9  | 91.1  | 90.2  | 88.9  | 85.4  | 78.1  | 69.1  | (28) |
|                                                                        | 2%          | WB     | 59.6  | 65.3  | 69.4  | 75.1  | 78.6  | 80.2  | 81.0  | 81.0  | 79.2  | 74.8  | 69.9  | 59.4  | (29) |
|                                                                        |             | MCDB   | 63.9  | 72.0  | 74.6  | 81.3  | 85.3  | 86.6  | 89.3  | 89.5  | 87.3  | 82.0  | 76.0  | 63.7  | (30) |
|                                                                        | 5%          | WB     | 55.5  | 62.5  | 66.5  | 73.6  | 77.5  | 79.4  | 80.2  | 80.1  | 78.3  | 73.0  | 67.1  | 56.2  | (31) |
|                                                                        |             | MCDB   | 59.9  | 67.8  | 70.8  | 79.1  | 83.1  | 85.6  | 88.2  | 88.4  | 86.1  | 79.6  | 72.0  | 61.5  | (32) |
|                                                                        | 10%         | WB     | 52.3  | 59.2  | 63.8  | 72.1  | 76.2  | 78.7  | 79.5  | 79.2  | 77.4  | 71.2  | 64.3  | 53.5  | (33) |
|                                                                        |             | MCDB   | 56.8  | 63.3  | 67.5  | 77.0  | 81.1  | 84.7  | 87.1  | 87.1  | 85.0  | 77.8  | 69.9  | 59.5  | (34) |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 10.5  | 10.7  | 10.9  | 12.2  | 12.9  | 11.9  | 14.0  | 14.8  | 15.5  | 15.6  | 14.9  | 13.7  | (35) |
|                                                                        | 5% DB       | MCDBR  | 20.3  | 20.0  | 19.4  | 18.1  | 18.8  | 16.1  | 18.0  | 18.9  | 19.5  | 21.2  | 22.7  | 23.2  | (36) |
|                                                                        |             | MCWBR  | 11.1  | 10.8  | 9.8   | 8.2   | 7.7   | 5.5   | 5.2   | 5.1   | 5.7   | 7.1   | 9.4   | 11.6  | (37) |
|                                                                        | 5% WB       | MCDBR  | 15.2  | 16.8  | 15.0  | 14.7  | 13.8  | 12.7  | 15.4  | 16.1  | 16.3  | 15.7  | 15.6  | 16.2  | (38) |
|                                                                        |             | MCWBR  | 10.1  | 10.3  | 10.0  | 8.5   | 7.1   | 5.3   | 5.5   | 5.3   | 5.9   | 7.2   | 8.8   | 10.1  | (39) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.647 | 0.682 | 0.799 | 0.764 | 0.633 | 0.602 | 0.568 | 0.637 | 0.668 | 0.714 | 0.618 | 0.626 | (40) |
|                                                                        | taud        |        | 1.587 | 1.568 | 1.409 | 1.511 | 1.799 | 1.888 | 1.976 | 1.779 | 1.689 | 1.551 | 1.698 | 1.644 | (41) |
|                                                                        | Ebn at Noon |        | 190   | 198   | 184   | 197   | 225   | 230   | 238   | 221   | 209   | 187   | 197   | 188   | (42) |
|                                                                        | Edh at Noon |        | 76    | 82    | 102   | 94    | 70    | 63    | 58    | 70    | 75    | 83    | 67    | 69    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 543   | 616   | 736   | 930   | 1112  | 1132  | 1480  | 1428  | 1281  | 1074  | 831   | 679   | (44) |
|                                                                        | RadStd      |        | 139   | 156   | 124   | 129   | 157   | 165   | 149   | 98    | 123   | 106   | 155   | 165   | (45) |

Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | +0.50 | N/A     | N/A    | +1.42   | N/A   | N/A   | N/A         | N/A   | +189  | +142  |
| Regional (6 neighbors) | +0.58 | N/A     | N/A    | +0.92   | N/A   | N/A   | N/A         | N/A   | +221  | +146  |

Nomenclature: See separate page